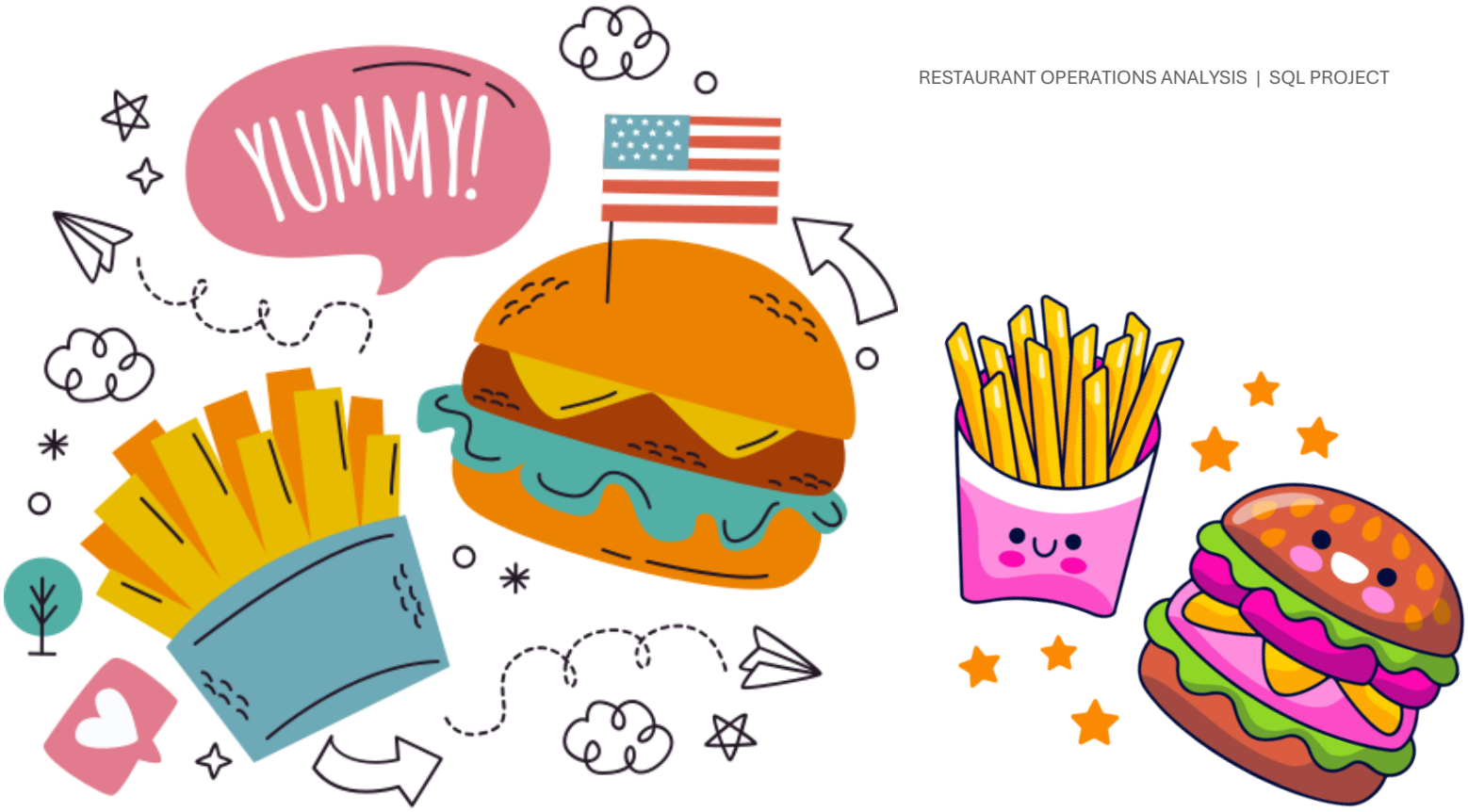


RESTAURANT OPERATIONS ANALYSIS

A SQL-Based Menu, Order, and Spending Analysis

PROJECT REPORT

Prepared using MySQL

Name : ABHINAV RAJ

1. Executive Summary

This project uses MySQL to examine restaurant menu data and order-detail records. The analysis covers menu composition and pricing, order activity during the first quarter of 2023, the most and least frequently ordered dishes, and the five orders with the highest calculated menu value.

The menu contains 32 items across four categories. Mexican and Italian each contain nine items, while Italian has the highest average menu price. The order-detail data records 5,370 distinct orders and 12,097 non-null item references between January 1 and March 31, 2023. The most frequently ordered item in the results is Hamburger (622 occurrences), and Chicken Tacos has the lowest count among the ten items returned by the least-frequently-ordered query (123 occurrences). Order 440 has the highest calculated menu value among the five highest-value orders at $192.15.

The findings are descriptive. Menu price totals are not confirmed final bills or profit, and order-item counts represent order-detail occurrences rather than unique customers.

2. Project Objectives

- Explore menu items and compare the number of dishes and prices across categories.
- Explore order details, including the date range, total orders, total item records, and large orders.
- Analyze menu-item ordering frequency and identify the five highest-spending orders using listed menu prices.

3. Dataset and Methodology

The analysis uses two MySQL tables: menu_items and order_details. The menu_items table contains menu_item_id, item_name, category, and price. The order_details table contains order_details_id, order_id, order_date, order_time, and item_id. The item_id field is matched to menu_item_id to attach menu names, categories, and prices to order records.

SQL methods used include aggregate functions (COUNT, MIN, MAX, AVG, SUM), GROUP BY, ORDER BY, LIMIT, HAVING, and INNER/LEFT JOIN operations. Results in this report are based on the query outputs shared during the project.

4. Objective 1 — Explore Menu Items

4.1 Menu overview

Metric	Result
Total menu items	32
Most expensive item	Shrimp Scampi — Italian — $19.95
Least expensive item	Edamame — Asian — $5.00
Italian dishes	9
Lowest-priced Italian dishes	Spaghetti — $14.50; Fettuccine Alfredo — $14.50
Highest-priced Italian dish	Shrimp Scampi — $19.95
Price range	$14.95

4.2 Menu items by category

Category	Number of Items	Average Price
Mexican	9	$11.80
Italian	9	$16.75

Asian	8	$13.48
American	6	$10.07

4.3 Findings and interpretation

- The menu contains 32 items distributed across four categories.
- Mexican and Italian are the largest categories, with nine items each. Together they account for 18 of 32 menu items (56.25%).
- Italian dishes have the highest average listed price ($16.75), while American dishes have the lowest average ($10.07).
- The menu price range is $14.95, from Edamame at $5.00 to Shrimp Scampi at $19.95.
- Menu prices alone do not establish sales performance, food cost, or profitability.

5. Objective 2 — Explore Order Details

5.1 Order activity summary

Metric	Result
First order date	January 1, 2023
Last order date	March 31, 2023
Reporting period	Q1 2023 (90 calendar days)
Distinct orders	5,370
Non-null item references	12,097
Average item references per order	Approximately 2.25
Maximum item count in a single order	14
Orders with more than 12 item references	20
Share of distinct orders with more than 12 items	Approximately 0.37%

5.2 Largest orders by item count

Order ID	Item Count
2675	14
443	14
1957	14
3473	14
330	14
440	14
4305	14
1274	13
2126	13
1734	13

5.3 Findings and interpretation

- The order records cover January 1 through March 31, 2023.
- There are 5,370 distinct order IDs and 12,097 non-null item_id references, averaging about 2.25 item references per order.
- Seven of the ten orders returned in the top-ten item-count query contain 14 items, the highest observed count in that result.
- Twenty orders contain more than 12 non-null item references, equivalent to approximately 0.37% of 5,370 distinct orders.

- The data does not explain why these orders were larger; possible explanations such as group dining should be treated as hypotheses, not findings.

6. Objective 3 — Analyze Ordering Patterns and High-Value Orders

6.1 Ten most frequently ordered menu items

Rank	Menu Item	Category	Times Ordered
1	Hamburger	American	622
2	Edamame	Asian	620
3	Korean Beef Bowl	Asian	588
4	Cheeseburger	American	583
5	French Fries	American	571
6	Tofu Pad Thai	Asian	562
7	Steak Torta	Mexican	489
8	Spaghetti & Meatballs	Italian	470
9	Mac & Cheese	American	463
10	Chips & Salsa	Mexican	461

6.2 Ten least frequently ordered menu items

Rank	Menu Item	Category	Times Ordered
1	Chicken Tacos	Mexican	123
2	Potstickers	Asian	205
3	Cheese Lasagna	Italian	207
4	Steak Tacos	Mexican	214
5	Cheese Quesadillas	Mexican	233
6	Chips & Guacamole	Mexican	237
7	Veggie Burger	American	238
8	Shrimp Scampi	Italian	239
9	Fettuccine Alfredo	Italian	249
10	Hot Dog	American	257

6.3 Five highest-spending orders

Rank	Order ID	Calculated Menu Value
1	440	$192.15
2	2075	$191.05
3	1957	$190.10
4	330	$189.70
5	2675	$185.10

6.4 Findings and interpretation

- Hamburger is the most frequently ordered item in the reported results, with 622 occurrences, closely followed by Edamame with 620.
- American and Asian dishes occupy six of the top ten positions combined, with three items from each category.
- Chicken Tacos has the lowest frequency among the ten items returned by the least-frequently-ordered query, with 123 occurrences. Potstickers is next at 205.
- Four Mexican dishes appear in the least-frequently-ordered list. This is a reason to investigate individual dishes, not evidence that the whole Mexican category performs poorly.
- Order 440 has the highest calculated menu value ($192.15), followed by order 2075 ($191.05). The difference between the first and fifth orders is $7.05.

- The high-value order query sums listed menu prices for each order. Taxes, discounts, service charges, and profitability were not established by the available results.

7. Business Recommendations

- Use the popularity results to guide further analysis of menu placement and promotion, especially for Hamburger, Edamame, Korean Beef Bowl, Cheeseburger, and French Fries.
- Investigate lower-frequency dishes such as Chicken Tacos by reviewing price, menu description, customer feedback, and preparation cost before deciding whether to change or remove them.
- Test meal bundles involving frequently ordered items only after checking whether those items are commonly purchased together in the same orders.
- Combine item frequency with menu price and food-cost data to estimate revenue and contribution margin; frequency alone is not a measure of profitability.
- Review the contents of the five highest-value orders in a future extension to identify which dishes contributed to their totals.

8. Limitations

- The report relies on the query outputs shared during the project; the raw dataset was not independently reprocessed for this report.
- Order-item counts are counts of item_id occurrences in order_details and do not identify unique customers.
- The high-value order totals are calculated from listed menu prices and may not equal final customer bills.
- No food cost, discount, tax, customer identifier, or profit data was provided in the reported results; therefore, profitability and customer-level behavior cannot be concluded.
- The least-frequently-ordered results are presented as the ten lowest counts reported by the query. The interpretation assumes the query output correctly represents the menu-item counts.

9. Conclusion

The Restaurant Operations Analysis project demonstrates how MySQL can be used to turn menu and order-detail records into practical descriptive insights. The analysis found a 32-item menu across four categories, 5,370 distinct orders during Q1 2023, and 12,097 non-null item references. Hamburger and Edamame were the most frequently ordered items in the supplied results, while Chicken Tacos had the lowest frequency among the ten least-frequently-ordered items. Order 440 had the highest calculated menu value among the five highest-value orders.

These findings provide a starting point for menu and operations decisions, but they should be combined with food costs, discounts, final bill values, and customer feedback before making pricing, inventory, or menu-removal decisions. The project also demonstrates practical use of SQL aggregation, grouping, sorting, filtering, and joins.

Appendix A — Representative SQL Queries

A1. Menu category counts and average prices

```
SELECT category,
       COUNT(*) AS total_dishes,
       ROUND(AVG(price), 2) AS average_price
FROM menu_items
GROUP BY category
ORDER BY total_dishes DESC;
```

A2. Order period

```
SELECT MIN(order_date) AS first_order_date,
       MAX(order_date) AS last_order_date
FROM order_details;
```

A3. Total orders and item references

```
SELECT COUNT(DISTINCT order_id) AS total_orders,
       COUNT(item_id) AS total_items_ordered
FROM order_details;
```

A4. Orders with more than 12 item references

```
SELECT COUNT(*) AS orders_with_more_than_12_items
FROM (
    SELECT order_id, COUNT(item_id) AS total_items
    FROM order_details
    GROUP BY order_id
    HAVING COUNT(item_id) > 12
) AS large_orders;
```

A5. Most frequently ordered menu items

```
SELECT mi.item_name,
       mi.category,
       COUNT(od.item_id) AS times_ordered
FROM order_details AS od
JOIN menu_items AS mi
  ON od.item_id = mi.menu_item_id
GROUP BY mi.menu_item_id, mi.item_name, mi.category
ORDER BY times_ordered DESC
LIMIT 10;
```

A6. Five highest-value orders by listed menu prices

```
SELECT od.order_id,
       ROUND(SUM(mi.price), 2) AS order_total
FROM order_details AS od
JOIN menu_items AS mi
  ON od.item_id = mi.menu_item_id
GROUP BY od.order_id
ORDER BY order_total DESC
LIMIT 5;
```

